Qualcomm Technologies, Inc.

Qualcomm Linux Memory Guide

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1 Memory overview

Qualcomm® chip products use low-power double data rate (LPDDR) dynamic random-access memory (DRAM) to support system operations. DRAM is a type of volatile memory that stores data frequently accessed by the processor and requires constant refreshing to retain information.

The Qualcomm memory controller supports JEDEC - compliant LPDDR4X, LPDDR5, and LPDDR5X DRAM components. You can select the type of DRAM based on your device requirements.

Note: See Hardware SoCs that are supported on Qualcomm® Linux®.

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QCS6490/QCS5430

The memory controller supports high-speed LPDDR4X and LPDDR5 SDRAM in dual-channel non-package-on-package (non-PoP) configurations.

DRAM capabilities of QCS6490 and QCS5430:

- LPDDR4X and LPDDR5 SDRAM supports a maximum memory density of 16 GB.
- LPDDR4X SDRAM operates at a clock speed of up to 2133 MHz clock speed (2 × 16-bit).
- LPDDR5 SDRAM operates at a clock speed of up to 3200 MHz clock speed (2 × 16-bit).

The following figure shows the DRAM memory channel architecture for QCS6490 and QCS5430:

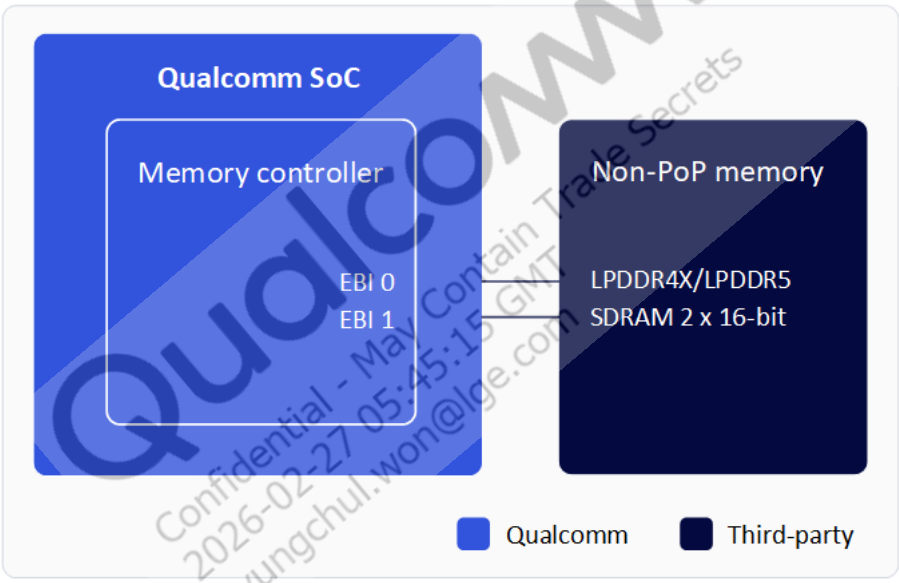

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Figure : QCS6490 and QCS5430 DRAM memory channel

IQ-9075

The memory controller supports LPDDR5X and LPDDR5 SDRAM in six-channel non-PoP configurations.

DRAM capabilities of Qualcomm Dragonwing™ IQ-9075:

- LPDDR5X and LPDDR5 SDRAM supports a maximum memory density of 36 GB.
- LPDDR5X and LPDDR5 SDRAM operates at a clock speed of up to 3200 MHz clock speed (6 × 16-bit).

The following figure shows the DRAM memory channel architecture for Qualcomm Dragonwing IQ-9075:

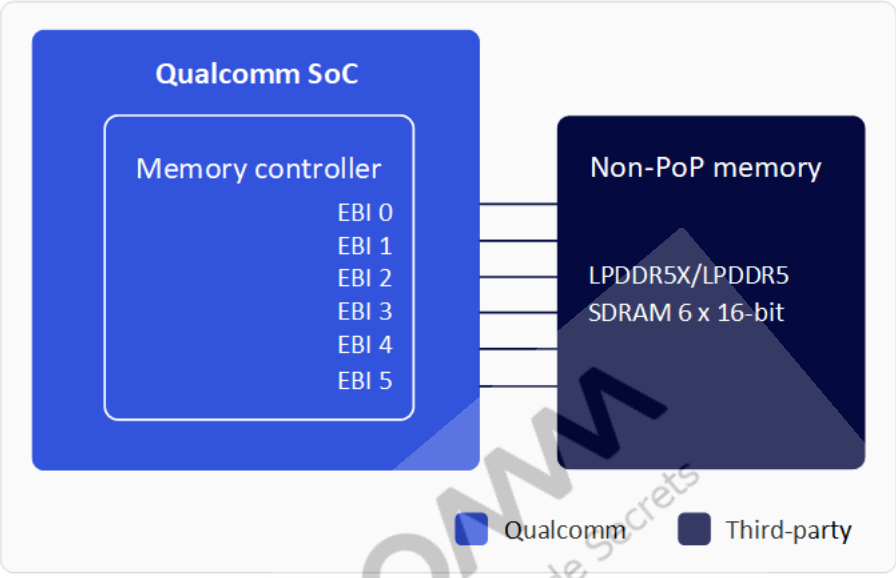

Figure : Dragonwing IQ-9075 DRAM memory channel

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IQ-8275

The memory controller supports LPDDR5X and LPDDR5 SDRAM in four-channel non-PoP configurations.

DRAM capabilities of Qualcomm Dragonwing™ IQ-8275:

- LPDDR5X and LPDDR5 SDRAM supports a maximum memory density of 32 GB.
- LPDDR5X and LPDDR5 SDRAM operates at a clock speed of up to 3200 MHz clock speed (4 × 16-bit).

The following figure shows the DRAM memory channel architecture for Qualcomm Dragonwing IQ-8275:

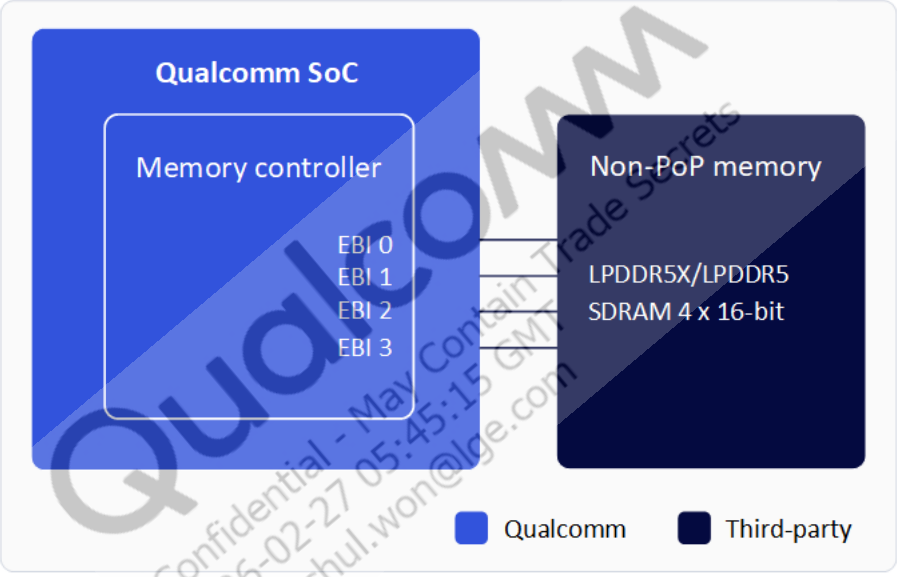

Figure : Dragonwing IQ-8275 DRAM memory channel

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IQ-615

The memory controller supports LPDDR4X SDRAM in dual-channel non-PoP configurations.

DRAM capabilities of Qualcomm Dragonwing™ IQ-615:

- LPDDR4X SDRAM supports a maximum memory density of 8 GB.
- LPDDR4X SDRAM operates at a clock speed of up to 1555.2 MHz clock speed (2 × 16-bit).

The following figure shows the DRAM memory channel architecture for Qualcomm Dragonwing IQ-615:

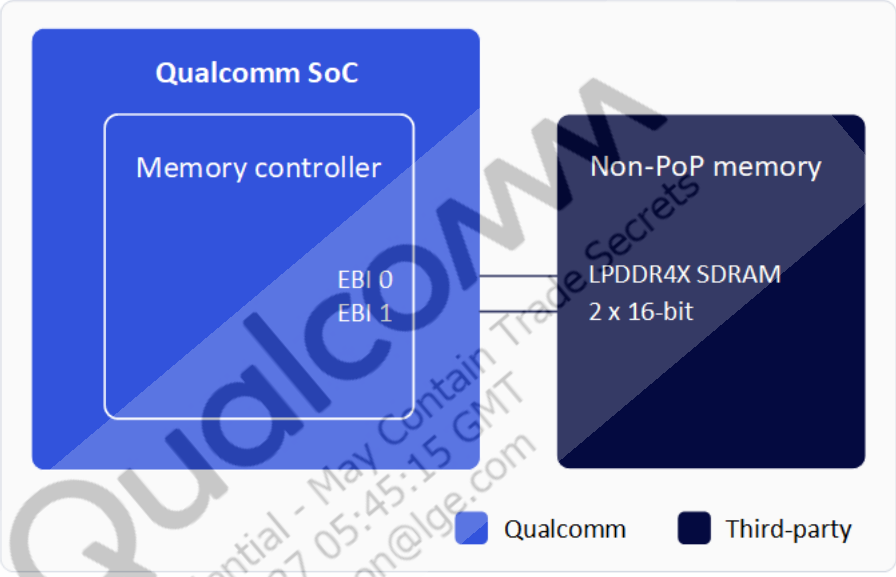

Figure : Dragonwing IQ-615 DRAM memory channel

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1.1 Enable DRAM

The DDR system firmware (DSF) is an embedded software code that allows the DRAM functionality. The eXtensible Boot Loader (XBL) image includes the DSF, which initializes the memory subsystem and runs during the boot sequence. The software works with all the DRAM components listed in the Qualcomm preferred vendor list (PVL). For information about using custom DRAM options from the PVL, see Customize and validate DRAM components.

The software doesn't require any extra modifications or configurations for the PVL DRAM components, because it automatically programs all required settings and timings during the bootup process.

1.2 Next steps

- Hardware SoCs
- eXtensible Boot Loader (XBL)

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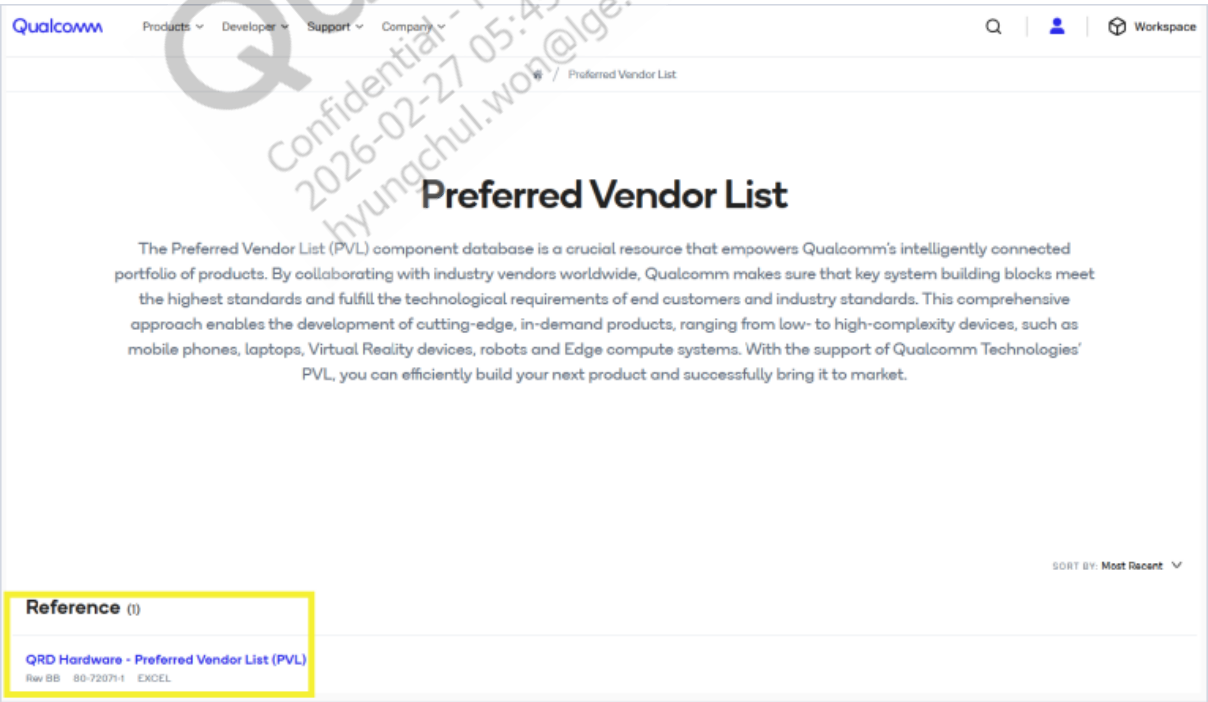
2 Customize and validate DRAM components

Qualcomm verifies and lists all DRAM components that work with its chip products in the preferred vendor list (PVL) component database.

Note: See Hardware SoCs that are supported on Qualcomm® Linux.®

To find DRAM components in the PVL, do the following:

1. Open the PVL component database.
2. Sign in to your account.
3. In the **Reference** section, select **QRD Hardware - Preferred Vendor List (PVL)** to download the Excel file.



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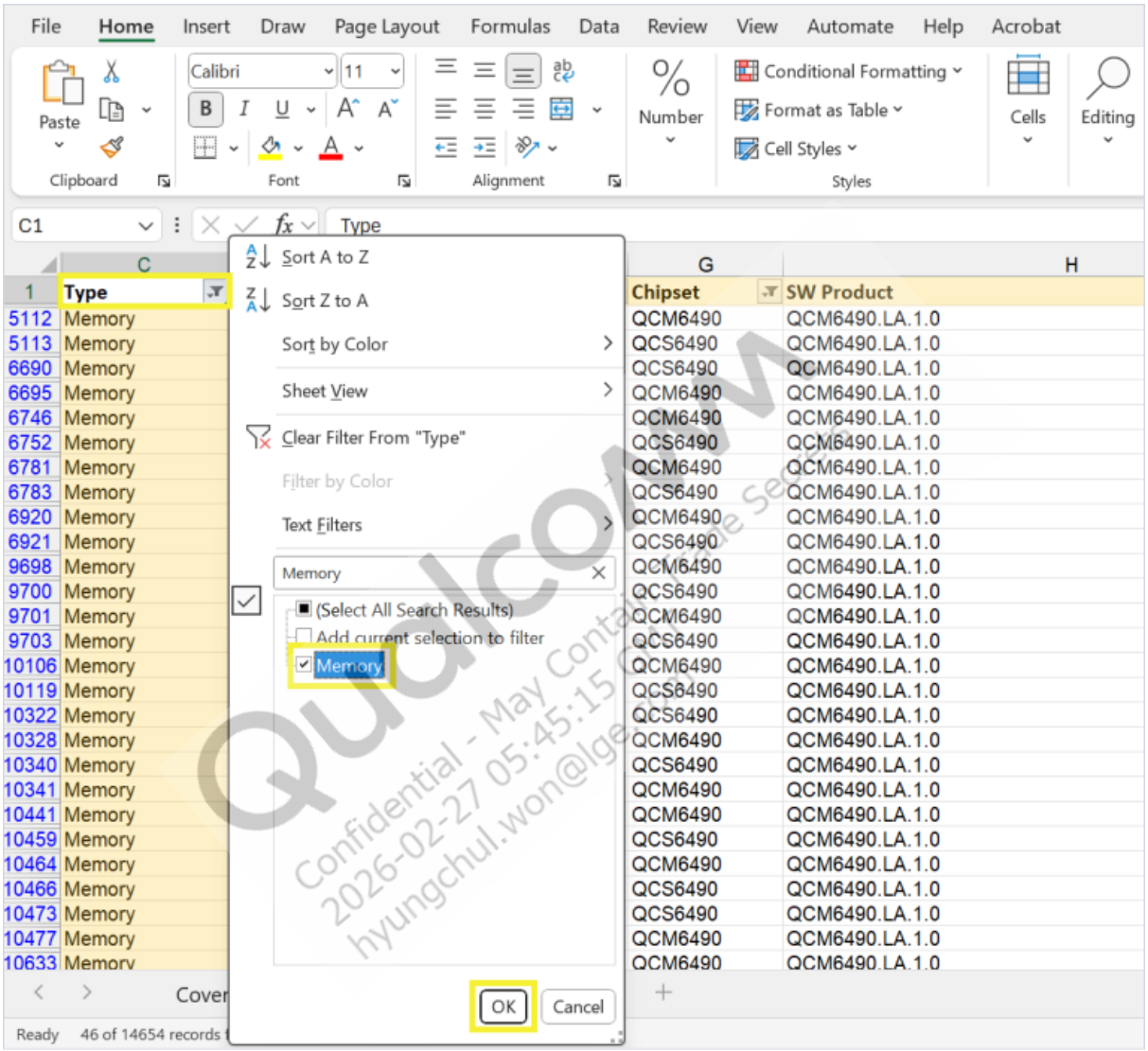
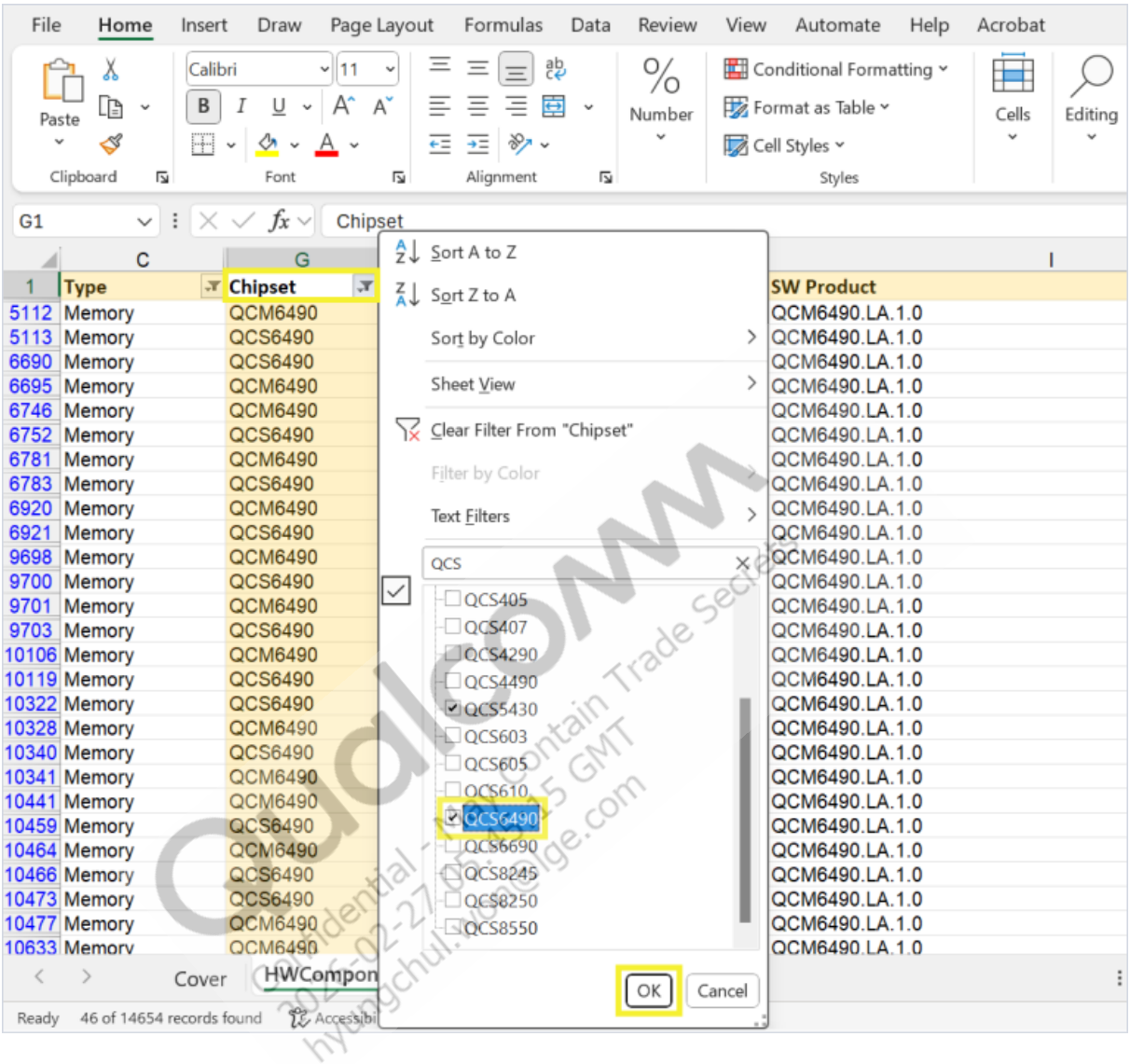
4. Open the downloaded Excel file and go to the **HWComponents** tab.
5. Apply an autofilter to the **Type** column, select **Memory**, and then select **OK** to confirm.



6. Choose a specific <Chipset> in the **Chipset** column and then select **OK** to apply the autofilter option. For example, replace <Chipset> with QCS6490.



If you intend to use a different DRAM component that's not listed in the PVL for your product, contact Qualcomm for software support.

Qualcomm reviews your validation request, authenticates the component, and provides the appropriate software. After validation, use the provided software and tools such as the Qualcomm memory stress application (QMESA). Use the QMESA tool to run a stress test and validate the DRAM.

2.1 Test DRAM using QMESA tool

QMESA is a performance-oriented cache and memory stress testing application used for conducting read and write stress tests specifically on DRAM.

2.2 Prerequisites for using the QMESA tool

Before setting up the QMESA tool, do the following:

- Use a Linux host computer to start the stress test.
- Download and install the Qualcomm® software center (QSC) GUI desktop application. For instructions, see Download QSC for Linux and Install QSC for Linux.
- Download QMESA for Linux Android/Linux Embedded from the QSC GUI desktop application. The QSC GUI desktop application provides QMESA as an executable binary (QMESA_64).

2.3 Run QMESA using Android debug bridge (ADB) interface

1. See Install and connect to ADB.
2. After the ADB interface configures, run QMESA using the following commands:

```
adb devices
adb root
adb push /<QMESA path>/QMESA_64 /tmp/
adb shell
root@<chipset>:~# cd /tmp/
root@<chipset>:/tmp# chmod 777 QMESA_64
root@<chipset>:/tmp# ./QMESA_64
```

2.4 Run QMESA using secure shell (SSH) interface

1. Set up a Wi-Fi connection and connect to the device through a SSH protocol interface. For instructions, see Sign in using SSH.
2. After the SSH interface configures, run QMESA using the following commands:

```
scp /<QMESA path>/QMESA_64 root@<remote host IP>:/tmp/
ssh root@<remote host IP>
root@<chipset>:~# cd /tmp/
root@<chipset>:/tmp# chmod 777 QMESA_64
root@<chipset>:/tmp# ./QMESA_64
```

Note:

- When prompted for a password, enter `<oelinux123>` to authenticate the file transfer through secure copy protocol (SCP).

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- Replace instances of `<chipset>` with the specific chipset applicable to you. For example, `qcs6490`.
- Run the QMESA test for 12-24 hours to achieve the appropriate stress results.

For more information about running QMESA, see the *Linux OS usage > QMESA usage* section in QMESA User Guide.

2.5 Next steps

- Download QSC for Linux
- Install QSC for Linux
- Install and connect to ADB
- Sign in using SSH

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3 Troubleshoot DRAM issues

The debug information guides you to check any issues during DRAM initialization from logs and monitor the DRAM usage information during runtime.

Qualcomm provides a sample log and an example command output to help with debugging.

Note: For any DRAM-related issues, such as initialization failures, kernel panics, bit flips, corruptions, or if you are working with board designs featuring new DRAM, contact Qualcomm for software support.

3.1 DRAM initialization

The DRAM initializes sequentially during the XBL stage of bootup.

For a sample boot UART log, see Troubleshoot using PBL/XBL log. The following table lists the sequence of DRAM initialization during bootup as indicated in the serial universal asynchronous receiver/transmitter (UART) logs.

DRAM initialization log entry and description

Log entry	Description
`sbl1_ddr_init, Start`	Starts DRAM initialization
`<LP4/LP5> DDR detected`	Detects a specific DRAM component
`Manufacturer ID = <XXXX>, Device Type = <XXXX>`	Provides the manufacturing ID and device type of the DRAM
`Rank 0 size = <XXXX> MB, Rank 1 size = <XXXX> MB`	Indicates the detection of the DRAM density
`Max enabled DDR Freq = <XXXX> MHz`	Shows the maximum DRAM clock frequency
`DDR - Image Load, Start`	Completes DRAM initialization and initiates image loading on the DRAM
`DDR Frequency, <XXXX> MHz`	Sets the DRAM clock frequency to `<XXXX>` MHz before exiting the XBL. After the XBL exits, you can adjust the DRAM frequency to meet the requirement.

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3.2 DRAM usage

To monitor real-time DRAM usage and available memory during runtime, run the following command:

```
cat /proc/meminfo
```

The following is a sample output:

```
cat /proc/meminfo
MemTotal: <XXXX> kB
MemFree: <XXXX> kB
MemAvailable: <XXXX> kB
```

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4 References

4.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
Qualcomm Linux Boot Guide	80-70023-4
Qualcomm Software Center User Guide	80-72780-2
QMESA User Guide	80-NH759-1
Resources	
JEDEC standard	https://www.jedec.org/category/technology-focus-area/mobile-memory-lpddr-wide-io-memory-mcp
Preferred vendor list	https://www.qualcomm.com/preferred-vendor-list
Qualcomm Software Center	https://softwarecenter.qualcomm.com/#/

4.2 Acronyms and terms

Acronym or term	Definition
ADB	Android debug bridge
DRAM	Dynamic random-access memory
DSF	DDR system firmware
LPDDR	Low-power double data rate
PoP	Package-on-package
PVL	Preferred vendor list
QMESA	Qualcomm memory stress application
QSC	Qualcomm Software Center
SCP	Secure copy protocol
SSH	Secure shell
UART	Universal asynchronous receiver/transmitter
XBL	eXtensible Boot Loader

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